A Project Report

On

"Samay – E-Commerce Website for Online Watch Shopping"

Under

E-Commerce

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by

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CERTIFICATE

This is to certify that the Project report entitled **"Samay – E-Commerce Website"** being submitted by **Samruddhi Adlinge (SIB47) And Manasi Kharche (SIB57)** is a record of bonafide work carried out by him/her under the supervision and guidance of **Dr. Deepali Ahir** in partial fulfillment of the requirement for **SE (Information Technology Engineering) – 2024 course** of Savitribai Phule Pune University, Pune in the academic year 2025-2026.

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ABSTRACT

This project presents the design and development of an e-commerce website named **SAMAY**, aimed at simplifying and enhancing the process of buying watches online. The primary objective of the system is to provide users with a convenient, fast, and user-friendly platform where they can browse, select, and purchase watches from anywhere at any time without the limitations of physical retail stores.

In today's digital era, traditional shopping methods for watches are often time-consuming and limited by location, availability, and store timings. The proposed system addresses these challenges by offering an efficient online solution that saves time, provides a wide variety of watch options—including men's, women's, digital, and sports watches—and improves accessibility for users. The platform is designed to ensure smooth interaction between users and the system through an intuitive and visually appealing interface.

The SAMAY website is developed using modern web technologies such as HTML, CSS, and JavaScript for designing an interactive frontend, while PHP is used for backend processing. MySQL database is used for storing user details, product information, and order data. The system follows a structured approach to handle user requests, maintain sessions, and ensure proper data management.

The key functionalities of the system include user registration and authentication, browsing and searching for watches, viewing product details, adding and removing items from the shopping cart, and placing orders. The application also maintains session-based cart management to ensure a seamless shopping experience. The system has been tested for functionality and performance, and it demonstrates reliable and efficient behavior under normal usage conditions.

Overall, this project highlights the practical implementation of an e-commerce system in the domain of online watch retailing. It shows how digital platforms can improve shopping experiences by offering convenience, variety, and accessibility. In the future, the system can be enhanced by integrating secure payment gateways, advanced search filters, recommendation systems, and mobile responsiveness to further improve user experience and system capabilities.

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CHAPTER 1

INTRODUCTION OF SAMAY

1.1 Overview

SAMAY is a web-based e-commerce platform developed to simplify the process of buying watches online. In today's fast-paced digital world, users prefer convenient and time-saving solutions for their daily needs, including purchasing watches. Traditional retail stores often have limitations such as restricted availability, limited stock, and dependency on physical presence. **SAMAY** addresses these challenges by providing a centralized digital platform where users can explore a wide variety of watches anytime and from anywhere..

The system allows users to browse different categories of watches such as men's, women's, digital, and sports watches, view detailed product descriptions, add items to their cart, and place orders easily. The platform is designed with a clean and user-friendly interface to ensure smooth navigation and an enhanced shopping experience.

The frontend of the system is developed using HTML, CSS, and JavaScript to provide an interactive user interface, while the backend is implemented using PHP to handle application logic. MySQL database is used to store user details, product information, and order data securely.

SAMAY aims to provide a simple, efficient, and reliable solution for online watch shopping while ensuring scalability for future enhancements such as payment integration, recommendation systems, and mobile application support.

1.1.1 Problem Statement

In the current scenario, traditional watch purchasing methods involve visiting physical retail stores, which can be time-consuming and inconvenient. Many users face difficulties in finding specific types of watches due to limited availability, restricted variety, or lack of proper inventory management. Additionally, comparing prices, exploring different styles such as men's, women's, digital, and sports watches, and accessing the latest designs becomes challenging in offline environments.

With the increasing demand for digital solutions, there is a need for an efficient platform that allows users to access a wide range of watches from a single location. Existing large-scale e-commerce platforms like Amazon and Flipkart provide such services, but they are often complex and overloaded with features, making them less suitable for beginner users or small-scale implementations.

Therefore, there is a need to develop a simple, lightweight, and user-friendly e-commerce system specifically focused on watch selling. SAMAY aims to solve this problem by providing an easy-to-use platform that simplifies product browsing, cart management, and order placement.

1.2 Aim

The main aim of the SAMAY project is to design and develop a user-friendly and efficient e-commerce website that allows users to browse, select, and purchase watches online with ease.
The project focuses on providing a smooth and simplified shopping experience while maintaining proper data management and system reliability.

1.3 Objectives

The primary objectives of the SAMAY are:

- To develop a secure user registration and login system
- To create an organized watch catalog for easy browsing
- To categorize watches into men's, women's, digital, and sports sections
- To implement a shopping cart system using session management
- To allow users to add and remove watches from the cart
- To enable users to place orders efficiently
- To store user, product, and order data in a structured database
- To provide a responsive and user-friendly interface

1.4 Scope

The scope of the SAMAY system is currently limited to an online platform for browsing and purchasing watches. It is designed for individual users such as fashion enthusiasts, students, professionals, and online shoppers.

The application can be accessed through web browsers on desktops and laptops. It provides basic e-commerce functionalities such as product listing, category-wise browsing, cart management, and order placement.

The future scope of the system includes:

- Integration of secure online payment gateways (UPI, PayPal, etc.)
- Development of a mobile-friendly or dedicated mobile application.
- Implementation of advanced search and filtering features (brand, price, type, etc.)
- Addition of recommendation systems based on user preferences.
- Expansion to include accessories such as straps, smartwatches, and other wearable products.

1.5 System Requirements and Specifications

1.5.1 Hardware Requirements

Component	Description
Device	Computer / Laptop
Processor	Minimum Dual Core
RAM	Minimum 4GB
Internet	Required for access

1.5.2 Software Requirements

Software	Description
Operating System	Windows / Linux / macOS
Development Tool	VS Code
Server	XAMPP (Apache + MySQL)
Backend	PHP
Frontend	HTML, CSS, JavaScript
Database	MySQL
Browser	Chrome, Edge, Firefox

CHAPTER 2
LITERATURE SURVEY

2.1 Introduction to Literature Survey

The literature survey focuses on studying existing e-commerce platforms, technologies, and features to understand how modern online shopping systems are designed and implemented. It helps in identifying the strengths, limitations, and key functionalities required for building an efficient e-commerce website like SAMAY.

2.2 Overview of E-Commerce.

E-commerce refers to the buying and selling of goods and services over the internet. It has transformed traditional business models by enabling online transactions and global accessibility.

Types of E-Commerce:

- B2C (Business to Consumer) – Example: Amazon
- B2B (Business to Business)
- C2C (Consumer to Consumer)

E-commerce plays a vital role in modern retail by offering convenience, variety, and accessibility.

2.3 Review of Existing Systems / Platforms

a. Amazon

- Features: Large product catalog, fast delivery, recommendations
- Strengths: Scalable system, advanced search
- Limitations: Complex UI for beginners

b. Flipkart

- Features: Indian market focus, offers, payment options
- Strengths: User-friendly interface
- Limitations: Heavy system with many features

c. Shopify

- Features: Online store creation
- Strengths: Easy setup
- Limitations: Paid platform

2.4 Technologies Used in E-Commerce

- Frontend: HTML, CSS, JavaScript
- Backend: PHP
- Database: MySQL
- Server: XAMPP
- Payment Gateway: PayPal (future scope)

2.5 Key Features Identified

- User authentication system
- Product catalog management
- Shopping cart system
- Order processing
- Database management

2.6 Challenges in Existing Systems

- Complex user interfaces
- High system dependency
- Security concerns
- Scalability issues

2.7 Summary of Findings

From the literature survey, it is observed that existing e-commerce platforms provide advanced features but are often complex and resource-heavy. SAMAY focuses on delivering a simplified and lightweight solution that retains essential functionalities while ensuring ease of use.

CHAPTER 3
SYSTEM ARCHITECTURE

3.1 Overview

SAMAY is designed as a web-based e-commerce application that follows a structured 3-tier architecture, consisting of the Frontend (User Interface), Backend (Application Logic), and Database (Data Storage). This layered architecture ensures proper separation of concerns, making the system easy to manage, scalable, and efficient.

The frontend layer acts as the visual interface through which users interact with the system. It allows users to browse watches, view product details, and perform actions such as adding items to the cart or placing orders. The frontend is developed using HTML, CSS, and JavaScript to ensure a responsive and user-friendly design.

The backend layer is responsible for handling all the business logic of the application. It processes user requests, manages sessions, handles form submissions, and interacts with the database. PHP is used as the backend language, as it integrates well with MySQL and is suitable for web-based applications.

The database layer stores all the important information such as user details, product catalog, and order records. MySQL is used as a relational database to ensure structured storage and efficient data retrieval.

Additionally, the system is hosted using the XAMPP server, which provides an Apache server environment along with MySQL support. This architecture ensures smooth communication between all components, resulting in a seamless online shopping experience.

3.2 Frontend

Technologies Used: HTML5, CSS3, JavaScript

The frontend of SAMAY is the part of the system that users directly interact with. It is designed to be simple, attractive, and easy to navigate, ensuring a smooth user experience.

Features of Frontend:

- Homepage: Displays available watches with images, names, and prices
- Navigation Bar: Provides links to Home, Categories (Men, Women, Digital, Sports), Cart, and Checkout pages
- Product Display: Shows watch details such as name, price, category, and image
- Cart Interface: Allows users to view selected items
- Responsive Design: Works on different screen sizes and devices

Working:

HTML is used to structure the web pages, CSS is used for styling and layout design, and JavaScript is used for adding interactivity such as button clicks and dynamic updates.

The frontend communicates with the backend using form submissions and HTTP requests.

3.3 Backend

Technology Used: PHP

The backend is the core part of the SAMAY system, responsible for handling logic, processing user actions, and managing communication between the frontend and database.

Functions of Backend:

- User Authentication: Handles user registration and login
- Session Management: Maintains user sessions and cart data
- Cart Operations: Adds/removes products from cart
- Order Processing: Handles checkout functionality
- Database Interaction: Fetches and stores data related to users, products, and orders

Working:

PHP scripts process requests sent from the frontend. For example:

- When a user clicks “Add to Cart”, PHP stores product data in session
- When a user visits the cart page, PHP retrieves session data
- During checkout, PHP processes the order and clears the cart

Security Features:

- Session handling to prevent unauthorized access
- Input validation for forms
- Basic protection against invalid data

3.4 Database

Technology Used: MySQL

The database plays a crucial role in storing and managing all application data. It ensures that user and product information is stored securely and retrieved efficiently.

1. **Users Table**
   Stores user account details.

Field	Description
id	Unique user ID (Primary Key)
username	Name of the user
email	User email
password	Encrypted password

2. **Products Table**
   Stores all watch information.

Field	Description
id	Unique product ID
name	Watch name
price	Price of watch
image	Watch image

3. **Orders Table**
   Stores order details.

Field	Description
id	Order ID
user_id	Linked to user
total_amount	Total order price
order_date	Date of order

Relationships Between Tables:

- **User → Orders (One-to-Many):**
  One user can place multiple orders
- **Products → Orders:**
  Products are included in orders

Conclusion of Architecture :

The SAMAY architecture is simple, efficient, and suitable for small-scale e-commerce systems. The separation of frontend, backend, and database ensures modular development and easy maintenance.

CHAPTER 4
RESULT AND DISCUSSION

Introduction

The “Results and Discussion” section describes the overall performance of the SAMAY system after implementation. It highlights how effectively the system meets its objectives, evaluates each feature, and analyzes system behavior under different conditions. This section not only presents the results but also interprets them to understand the strengths and limitations of the developed system.

4.1 Functional Results (System Working)

This section explains how each feature of the system performed during execution.

4.1.1 User Registration & Login

- **Expected Result:**
  Users should be able to register with valid details and login securely.
- **Actual Result:**
  The system successfully stores user data in the MySQL database and allows users to login using correct credentials.
- **Status:**
  Working successfully

4.1.2 Product Search and Filtering

- **Expected Result:**
  Users should be able to view and search watches easily.
- **Actual Result:**
  The system displays available watches properly across categories such as men’s, women’s, digital, and sports watches. Basic browsing works well, but advanced filtering options are limited.
- **Status:**
  Working (basic level)

4.1.3 Add to Cart & Remove from Cart

- **Expected Result:**
  Users should be able to add/remove books dynamically.

- **Actual Result:**

  Products are stored using session variables. Users can add and remove items without any issues.

- **Status:**

  Working successfully

4.1.4 Checkout Process

- **Expected Result:**

  Users should be able to place orders after reviewing the cart.

- **Actual Result:**

  Checkout page confirms order placement and clears the cart successfully.

- **Status:**

  Working successfully

4.1.5 Payment Integration

- **Expected Result:**

  Secure online payment should be processed.

- **Actual Result:**

  A basic/dummy payment system (PayPal sandbox) is implemented for demonstration purposes.

- **Status:**

  Partially implemented

4.1.6 Admin Panel

- **Expected Result:**

  Admin should manage watch products.

- **Actual Result:**

  Basic product handling (adding, updating, deleting watches) is possible, but full admin dashboard features are limited.

- **Status:**

  Limited functionality

4.2 Output Screens / Snapshots

The system interface of SAMAY was tested and verified using different pages of the application. Each page plays an important role in providing a smooth and user-friendly experience.

- **Login Page**

The Login Page allows existing users to access their accounts securely by entering their registered email and password. The system validates user credentials using backend authentication.

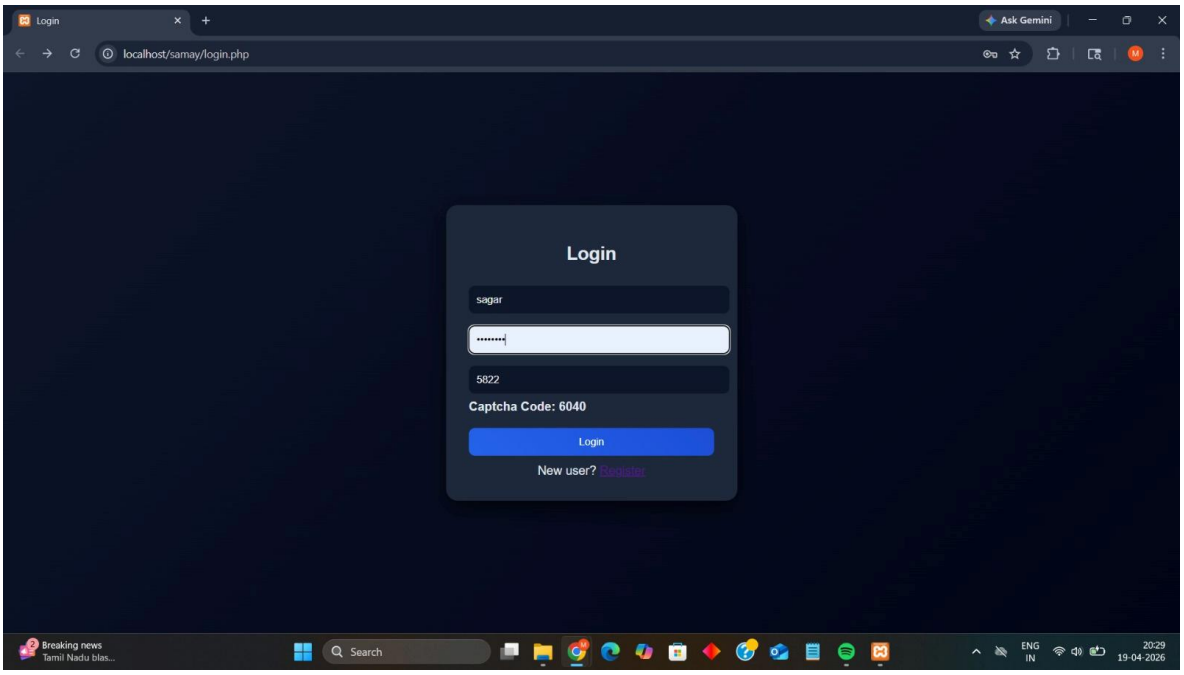


Fig 5.1 Login Page

- **Registration Page**

The Registration Page is used by new users to create an account. Users are required to enter details such as username, email, and password. After successful registration, the data is stored in the database.

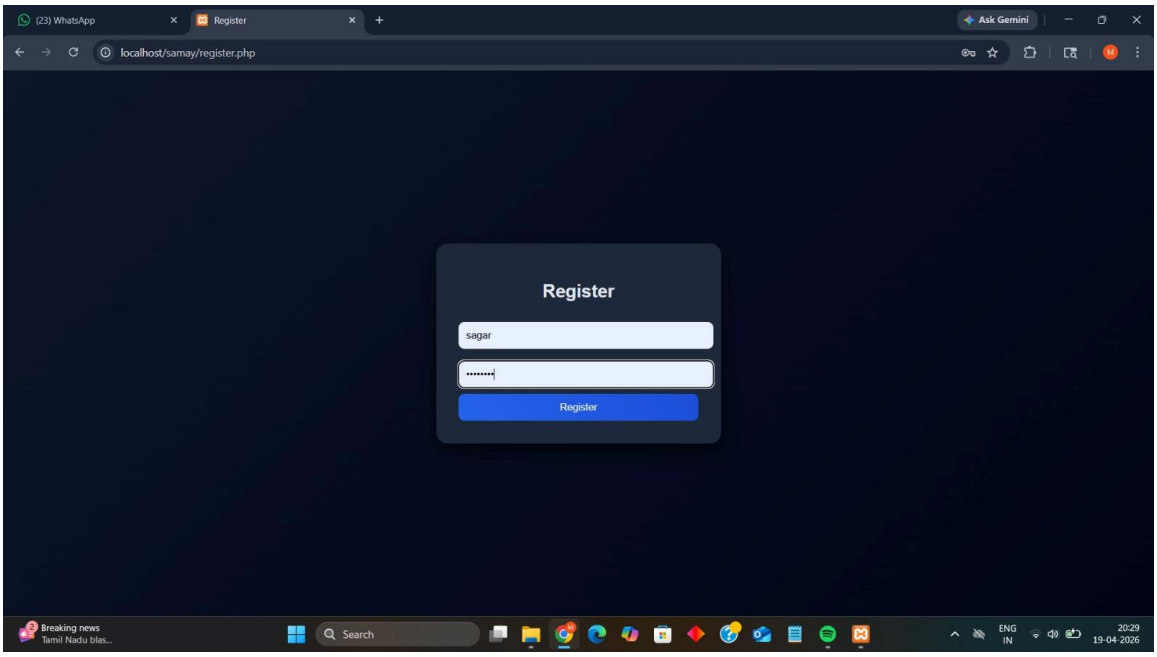


Fig 5.2 Registration Page

- **Homepage**

The Homepage displays the list of available watches along with their images, descriptions, and prices. It acts as the main interface where users can browse products.

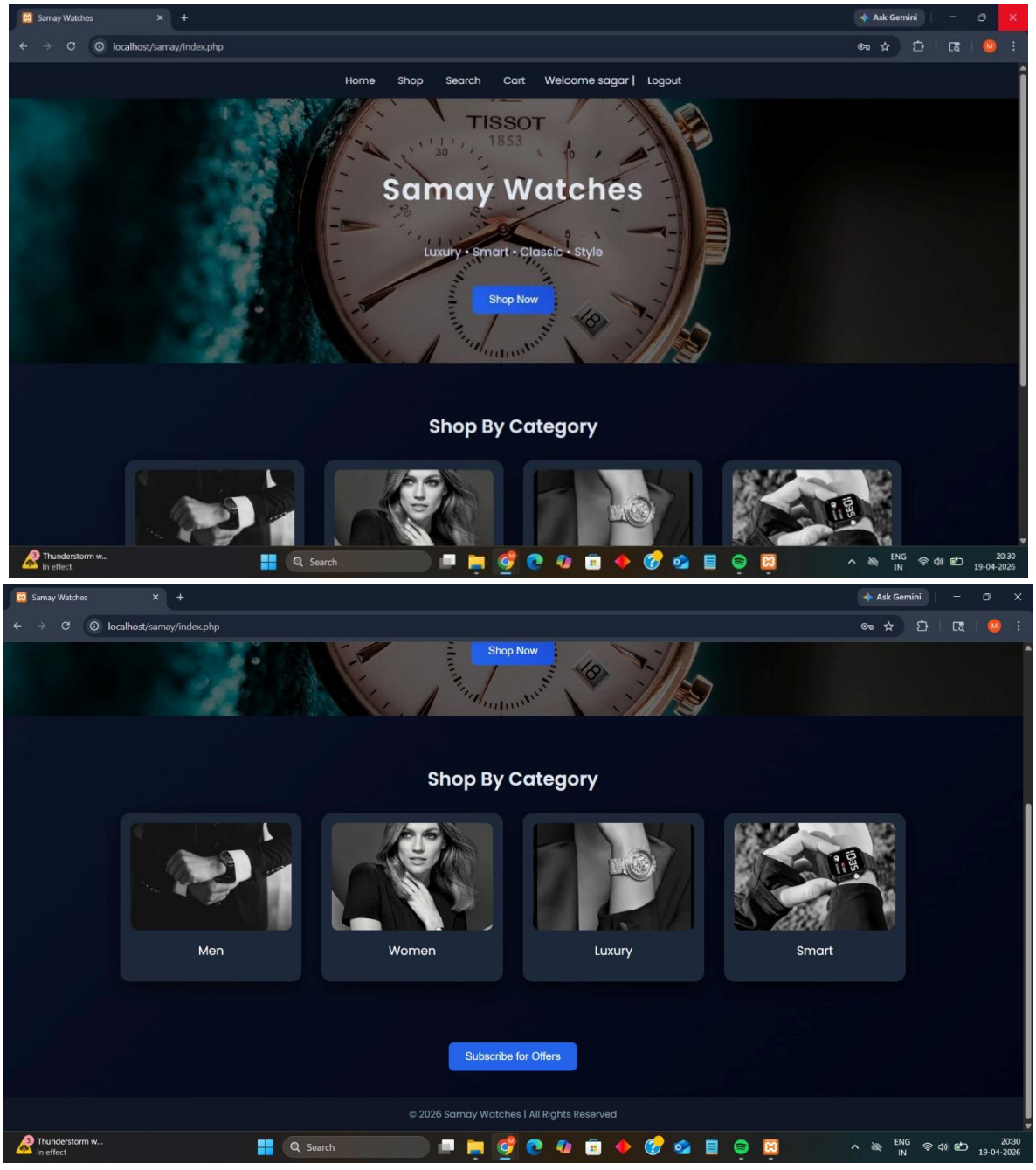
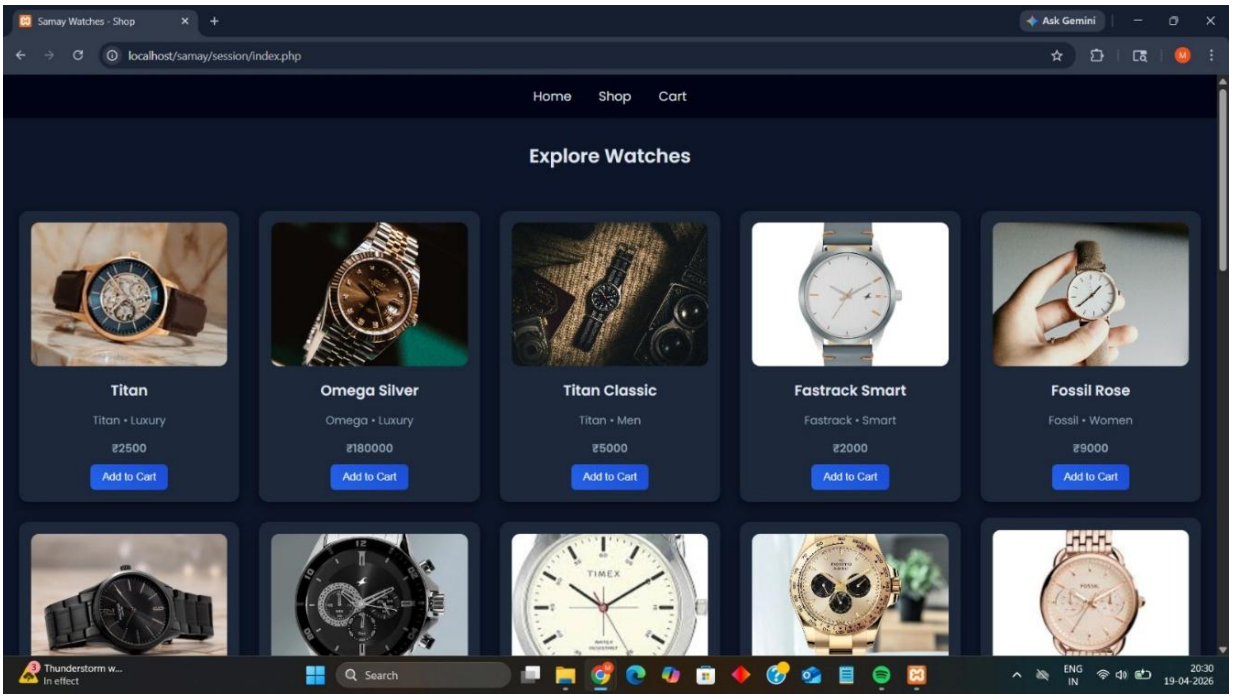
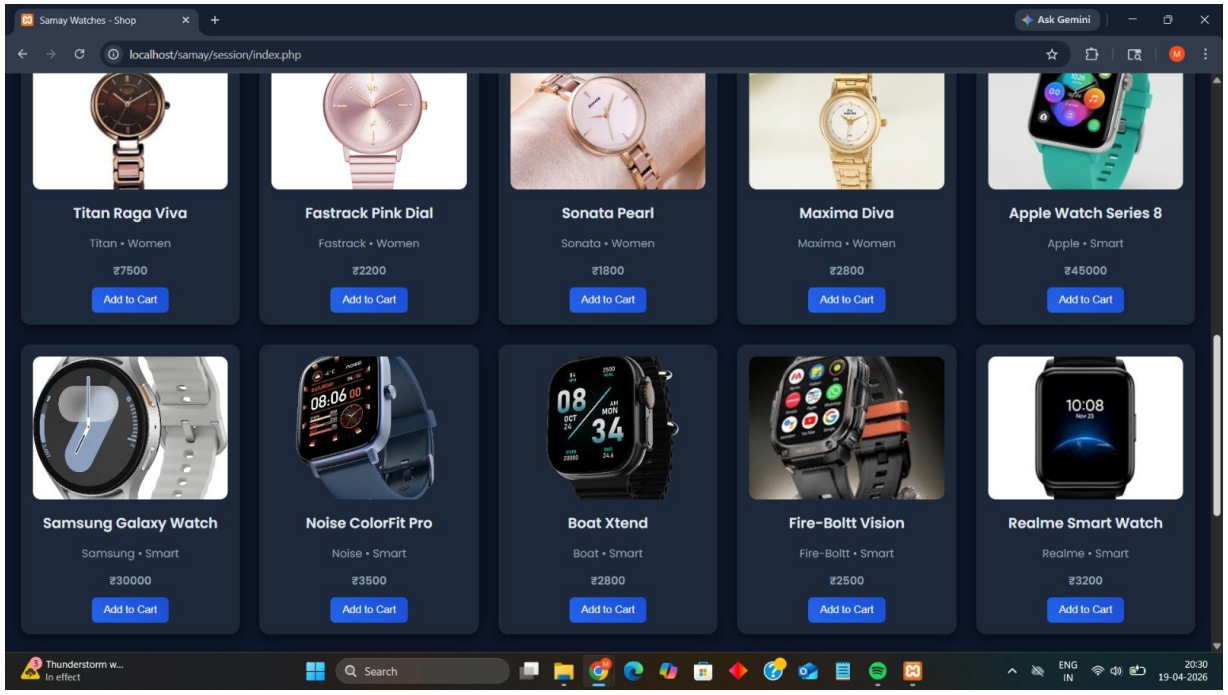


Fig 5.3 Homepage

- **Product Page**

The Product Page shows detailed information about a selected watch, including its name, price, and description. Users can add the product to the cart from this page.





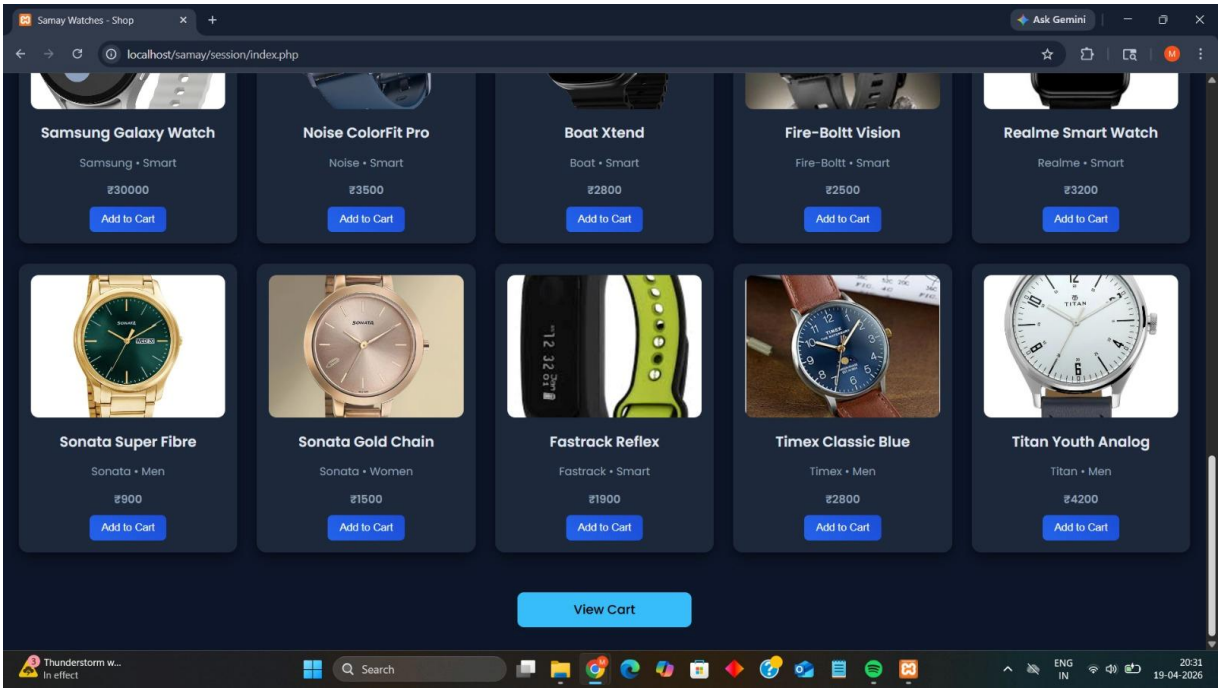


Fig 5.4 Product Page

- **Cart Page**

The Cart Page displays all the selected products along with their quantity and total price. Users can also remove items from the cart.

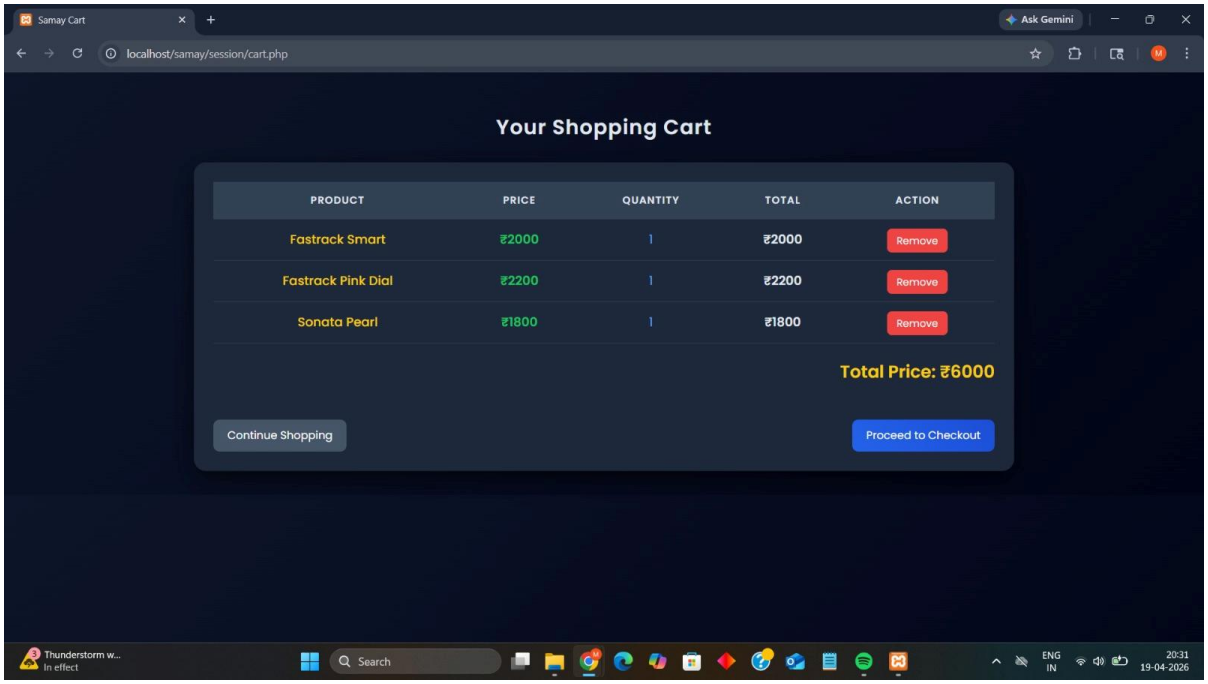


Fig 5.5 Cart Page

- **Checkout Page**

The Checkout Page confirms the order placement. After checkout, the system processes the order and clears the cart.

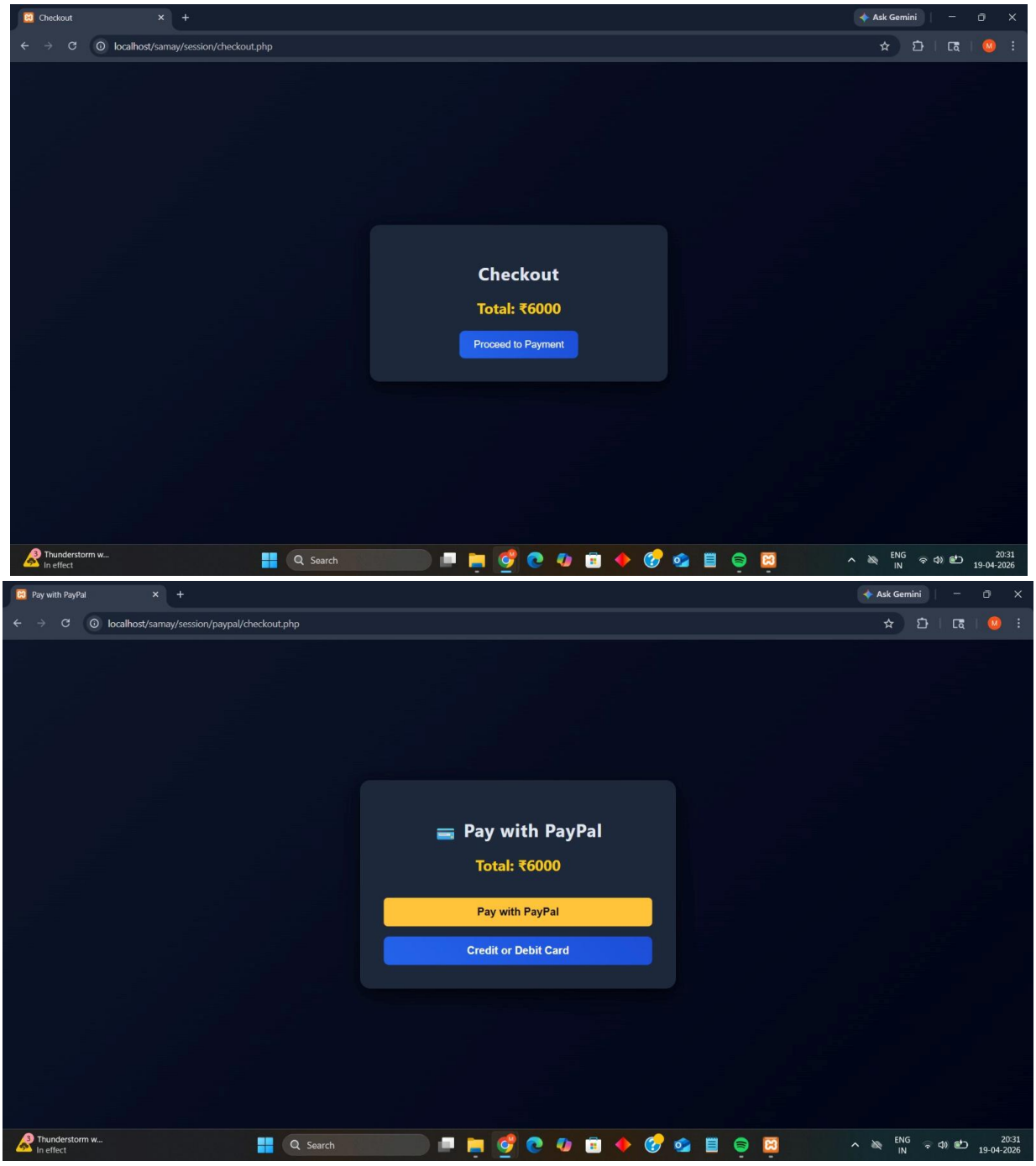


Fig 5.6 Checkout Page

- **Premium Subscription**

The Premium Subscription feature in SAMAY allows users to access exclusive benefits such as special discounts on watches, early access to new collections, and premium customer support. This feature enhances user experience by providing added value and encourages customer retention. It also helps in improving user engagement and offers a more personalized shopping experience.

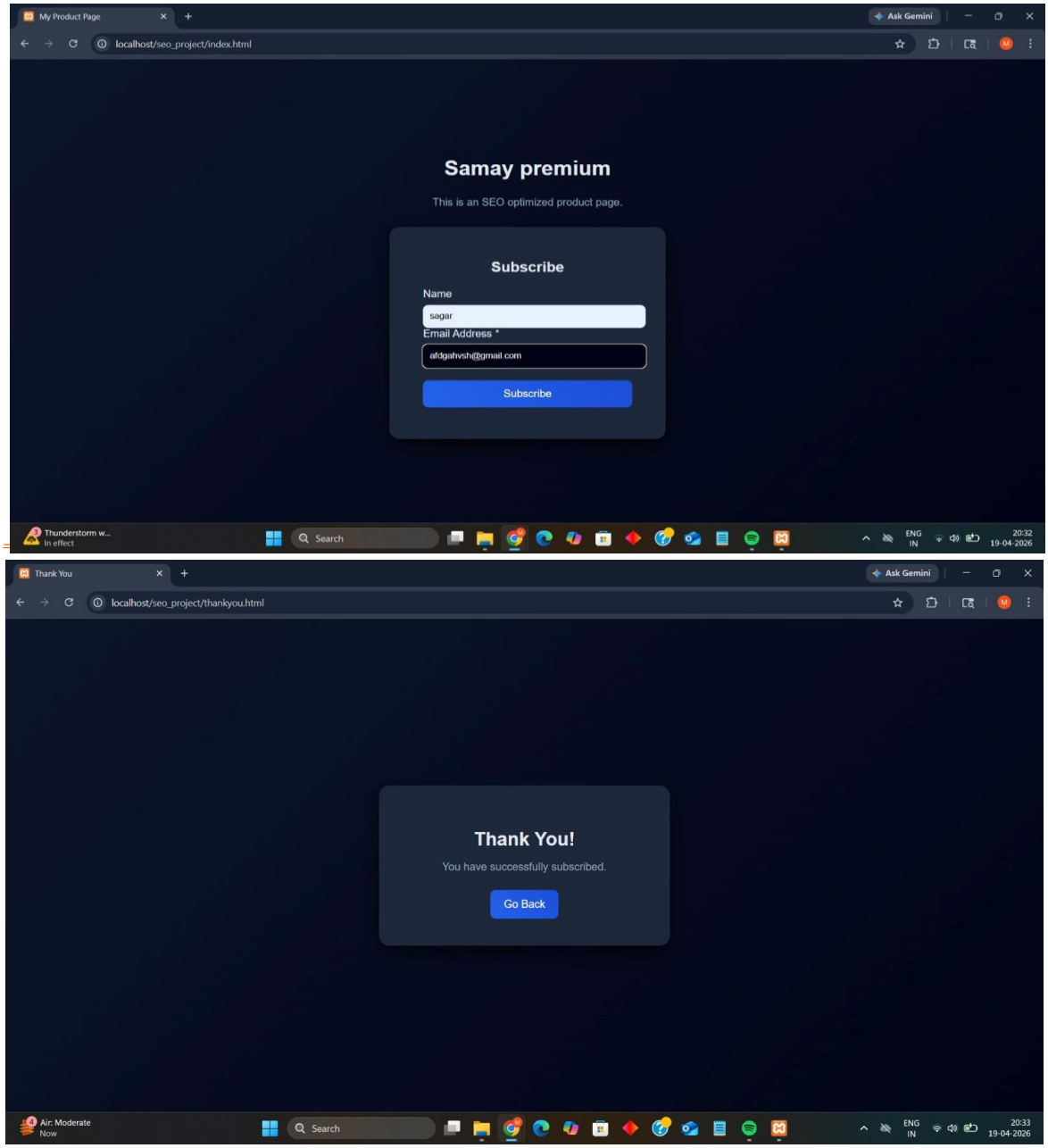


Fig 5.7 Premium Subscription Page

4.3 Performance Analysis

The performance of the SAMAY was evaluated based on several factors:

- **Page Loading Speed:**

  Pages load quickly under normal conditions due to simple design and lightweight structure.

- **System Response Time:**

  The system responds instantly to user actions like adding items to the cart and navigating pages.

- **Handling Multiple Users:**

  The system performs well for small-scale usage. However, it is not optimized for high traffic or concurrent users.

- **Issues Observed:**
  - Slight delay when loading multiple images
  - No caching mechanism implemented

Overall, the system shows satisfactory performance for a mini-project level implementation.

4.4 Testing Results

Different types of testing were performed to ensure system reliability.

1. **Unit Testing**
Individual components like login, cart, and checkout were tested separately.
2. **Integration Testing**
Frontend and backend interactions were tested together.
3. **System Testing**
Complete system testing was done to verify overall functionality.

Test Case Table

Test Case	Expected Result	Actual Result	Status
User Login	Successful login	Working	Pass
Add to Cart	Item added	Working	Pass
Remove Item	Item removed	Working	Pass
Checkout	Order placed	Working	Pass
Database Storage	Data saved	Working	Pass

4.5 Discussion

The SAMAY system performs efficiently for its intended purpose of online watch shopping.

What Worked Well:

- Simple and user-friendly interface
- Smooth navigation between pages
- Fast cart operations
- Easy checkout process
- Clear categorization of watches (men's, women's, digital, sports)

Issues Identified:

- Lack of advanced filtering/search features (e.g., brand, price range)
- No real payment gateway integration
- Limited scalability for handling large datasets or users

Comparison with Existing Systems:

Compared to platforms like Amazon and Flipkart, the SAMAY system is much simpler and lightweight. While it lacks advanced features, it is easier to use and understand, making it suitable for academic purposes and small-scale implementations.

4.6 Limitations of the System

Although the system works successfully, it has some limitations:

- No real payment gateway (only demo/sandbox implementation)
- Limited product database (restricted number of watches)
- Basic user interface design
- No recommendation system based on user preferences
- Not optimized for large-scale users or high traffic

CHAPTER 5
APPLICATION AND TARGET USERS

5.1 Application

SAMAY is a web-based e-commerce platform designed to simplify and modernize the process of purchasing watches. It provides users with a centralized digital system where they can browse a wide variety of watches, compare options, and place orders conveniently from any location. Unlike traditional retail stores, which require physical presence and are limited by time and stock availability, SAMAY enables users to access watches anytime and anywhere through an internet-enabled device.

The application focuses on providing a smooth and user-friendly shopping experience by integrating essential e-commerce features such as product browsing, category-wise exploration (men's, women's, digital, sports), cart management, and order placement. The system ensures that users can easily navigate between different sections like homepage, product details, cart, and checkout without confusion.

SAMAY transforms the traditional shopping approach into a digital experience by reducing manual effort and saving time. It also helps in maintaining organized records of users, products, and orders through a structured database system. The use of PHP and MySQL ensures efficient backend processing and secure data handling.

The platform can also be extended in the future to include advanced features such as secure payment gateways, recommendation systems, and mobile compatibility. Overall, SAMAY demonstrates how digital solutions can enhance the accessibility and efficiency of watch purchasing systems.

5.2 Target Users

The SAMAY system is designed for a wide range of users who are interested in purchasing watches online. The main target users include:

1. **Fashion Enthusiasts**

   Individuals who are interested in styling and accessories can explore different types of watches to match their outfits and preferences. SAMAY provides them with a variety of trendy options.

2. **Professionals**

Working professionals often look for formal and stylish watches. The platform helps them easily find suitable watches without visiting multiple stores.

3. **Sports and Fitness Users**

Users who prefer sports or digital watches for fitness tracking or durability can explore relevant categories easily on the platform.

4. **Online Shoppers**

Users who prefer online shopping due to convenience and time-saving benefits can use SAMAY as an efficient alternative to traditional shopping methods. It allows them to browse and buy watches with minimal effort.

5. **Watch Enthusiasts**

People who have a strong interest in watches and collections can benefit from the platform by accessing a variety of styles and categories in one place. It helps them stay updated with new trends and manage their purchases easily.

CHAPTER 6
CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The SAMAY project successfully demonstrates the design and implementation of a basic e-commerce system for online watch shopping. The system provides essential functionalities such as user registration and login, product browsing, category-wise exploration (men's, women's, digital, sports), shopping cart ma nagement, and order placement. These features work together to deliver a smooth and efficient user experience.

The application replaces traditional watch purchasing methods with a digital solution that saves time, improves accessibility, and simplifies the buying process. Users can easily explore watches, manage their cart, and complete orders without any technical complexity.

The system is developed using HTML, CSS, and JavaScript for the frontend, and PHP with MySQL for backend processing and data storage. This combination ensures proper interaction between the user interface and the database, resulting in reliable performance.

Although the system is simple compared to large-scale platforms like Amazon and Flipkart, it effectively fulfills the objectives of a mini-project by providing a clear understanding of how e-commerce systems function.

Overall, the SAMAY project serves as a strong foundation for learning web development concepts and demonstrates the practical implementation of an online watch shopping system.

6.2 Future Scope

The SAMAY system can be further enhanced by adding advanced features and improving its scalability. Some of the possible future improvements are:

1. Online Payment Integration

Currently, the system does not support real payment processing. In future, secure payment gateways such as PayPal, UPI, or credit/debit cards can be integrated to allow real-time transactions.

2. Mobile Application Development

A mobile-friendly version or a dedicated mobile application can be developed to improve accessibility and allow users to shop using smartphones.

3. AI-Based Recommendation System

Artificial Intelligence can be used to recommend watches based on user preferences, purchase history, and browsing behavior, enhancing user experience.

4. Advanced Search and Filtering

The system can be improved by adding filters such as price range, brand, category, and ratings to make product searching more efficient.

5. Admin Dashboard Enhancement
A complete admin panel can be developed for managing products, users, and orders efficiently with proper analytics and reporting tools.

6. Scalability and Security Improvements

- Implementation of secure authentication (password hashing, encryption)
- Ability to handle large number of users
- Database optimization

7. Multi-Category Expansion
The platform can be expanded beyond watches to include related accessories such as straps, smartwatches, and other wearable products, making it a complete e-commerce platform.